

Conversion from INVICAT fluxes to EnviFlux fluxes

July 31, 2025

INVICAT and Enviflux fluxes have the following units:

- INVICAT fluxes, χ_{INV} , are in units of $\text{molecules cm}^{-2} \text{s}^{-1}$,
- Enviflux fluxes, χ_{Env} , are in units of $10^9 \text{kg m}^{-2} \text{s}^{-1}$ (change in methane mass per square metre per second multiplied by 10^9). Alternatively χ_{Env} are in units of $\mu\text{g m}^{-2} \text{s}^{-1}$.

Define the following symbols:

- $\delta\#\text{CH}_4$ is the number of methane molecules introduced by the fluxes in the surface grid box per unit time,
- $\#\text{air}$ is the reference number of air molecules in the surface grid box,
- Z_1 is the thickness of the surface grid box (m),
- A is an area of interest of the surface (m^2),
- N_A is Avogadro's constant $N_A = 6.022 \times 10^{23}$,
- ρ is the surface air density $\rho = 1.255 \text{kg m}^{-3}$,
- Δt is the time interval of interest (s),
- M_d is the dry air molecular weight $M_d = 28.97 \text{g mol}^{-1}$.

The fractional change in the number of molecules in the surface layer is

$$\delta x = \frac{\delta\#\text{CH}_4 \Delta t}{\#\text{air}} 10^9, \quad (1)$$

(the 10^9 is to put the quantity into ppb). $\delta\#\text{CH}_4$ can be found from χ_{INV} :

$$\delta\#\text{CH}_4 = 10^4 \chi_{\text{INV}} A. \quad (2)$$

$\#\text{air}$ can be calculated:

$$\#\text{air} = \frac{\rho Z_1 A}{M_d \times 10^{-3} / N_A}. \quad (3)$$

Substituting (2) and (3) into (1):

$$\begin{aligned} \delta x &= 10^4 \chi_{\text{INV}} A \Delta t 10^9 \times \frac{M_d \times 10^{-3} / N_A}{\rho Z_1 A} \\ &= \frac{10^{10} \chi_{\text{INV}} M_d \Delta t}{\rho Z_1 N_A}. \end{aligned} \quad (4)$$

In EnviFlux, the update has the following form:

$$\delta x = \frac{\Delta t}{\rho Z_1} \chi_{\text{Env}}. \quad (5)$$

We are looking for the value of χ_{Env} . Merging (4) with (5) gives

$$\begin{aligned} \frac{10^{10} \chi_{\text{INV}} M_d \Delta t}{\rho Z_1 N_A} &= \frac{\Delta t}{\rho Z_1} \chi_{\text{Env}} \\ \frac{\chi_{\text{INV}} M_d 10^{10}}{N_A} &= \chi_{\text{Env}}. \end{aligned} \quad (6)$$

The maximum value of χ_{INV} is about $2.7 \times 10^{12} \text{molecules cm}^{-2} \text{s}^{-1}$. This makes the maximum value of χ_{Env} around

$$\chi_{\text{Env}} = \frac{2.7 \times 10^{12} \times 28.97 \times 10^{10}}{6.022 \times 10^{23}} = \frac{2.7 \times 28.97 \times 10^{22}}{6.022 \times 10^{23}} = \frac{2.7 \times 28.97 \times 10^{-1}}{6.022} \approx 1.3 \text{ppbs}^{-1}$$

When this value is used inside EnviFlux, the maximum grid-point methane values increase to between 20000 and 70000 ppb. This is likely because there is no chemistry scheme in EnviFlux.